178057: biotite monzogranite, Kurrajong Well

Location and sampling

SANDSTONE (SG 50-16), LAKE MASON (2842)
MGA Zone 50, 750960E 6911380N

Sampled on 30 October 2002

The sample was collected from the edge of a low rocky pavement, 150 m southeast of the Sandstone–Wiluna Road and 7 km southwest of Kurrajong Well.

Tectonic unit/relations

The sample is a pale-grey, medium- and even-grained, massive biotite monzogranite from the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Wyche, 2004). It was collected to determine the age of monzogranite away from major shear zones.

Petrographic description

The principal minerals are plagioclase (35–40 vol.%), K-feldspar (35 vol.%), quartz (25 vol.%), biotite (2–3 vol.%), and opaque oxide (1 vol.%), with trace amounts of accessory apatite, chlorite, epidote, stilpnomelane, sericite, clinozoisite, allanite, and zircon. Anhedral to subhedral plagioclase from 0.4 to 3 mm in grain size could be andesine–oligoclase. Anhedral microcline is also abundant, having a similar grain size to the plagioclase, but the quartz occurs as partly poikilitic interstitial grains up to 4 mm long. Biotite, from 0.2 to 2 mm in grain size, is mostly aligned and there is some suggestion of an S–C fabric. However, as the other minerals show no evidence of deformation, this could be a primary magmatic foliation. Disseminated grains and lenses of opaque oxide are up to 0.8 mm long, and there is also disseminated apatite and altered allanite. Some of the smaller zircons, less than 70 µm in length, are clouded and fractured, but others, from 40 to 100 µm in diameter, are clear and partly zoned. Coarse-grained poikilitic muscovite seen in some of the plagioclase could have a deuteric or low-temperature metamorphic origin, with chlorite, epidote or both replacing biotite. Rare ?ferristilpnomelane suggests low greenschist-facies metamorphism. Irregular patches of sericite and clouded clinozoisite indicate weak to moderate, low-temperature hydrothermal, saussurite-like alteration. Narrow carbonate-filled fractures are a few micrometres wide. The fabric in this sample is essentially unmetamorphosed and undeformed, with some low-temperature alteration producing chlorite–epidote–stilpnomelane assemblages under low greenschist-facies conditions, and weak to moderate saussuritization of plagioclase.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown and black, euhedral and slightly elongate, and between 50 × 80 µm and 160 × 80 µm in size. A majority of grains have faint internal zonation, whereas the remainder are internally structureless. Many grains have a speckled appearance in transmitted light, and many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 13 and 17 April 2003. The counter deadtime during both analysis sessions was 24 ns. During the first analysis sesssion, ten analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.63% (1σ). Analyses 1.1 to 13.1 were obtained during the first analysis session. During the second analysis session, five analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.74% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 1.1, 3.1, 9.1, 18.1, 20.1, and 25.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-five analyses were obtained from 25 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to slightly discordant, with the discordance pattern consistent with ancient and recent episodes of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, many analyses can be assigned to one of three groups. Nine concordant to slightly discordant analyses of nine zircons (2.1, 4.1, 11.1, 12.1, 14.1, 17.1, 19.1, 22.1, 24.1), assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2648 ± 6 Ma (chi-squared = 1.50). Six concordant to slightly discordant analyses of six zircons (6.1, 8.1, 9.1, 21.1, 23.1, 25.1) have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2708 ± 9 Ma (chi-squared = 1.29). Concordant to slightly discordant analyses 5.1, 15.1, and 20.1, assigned to Group 3, have $^{207}Pb/^{206}Pb$ ratios defining a single

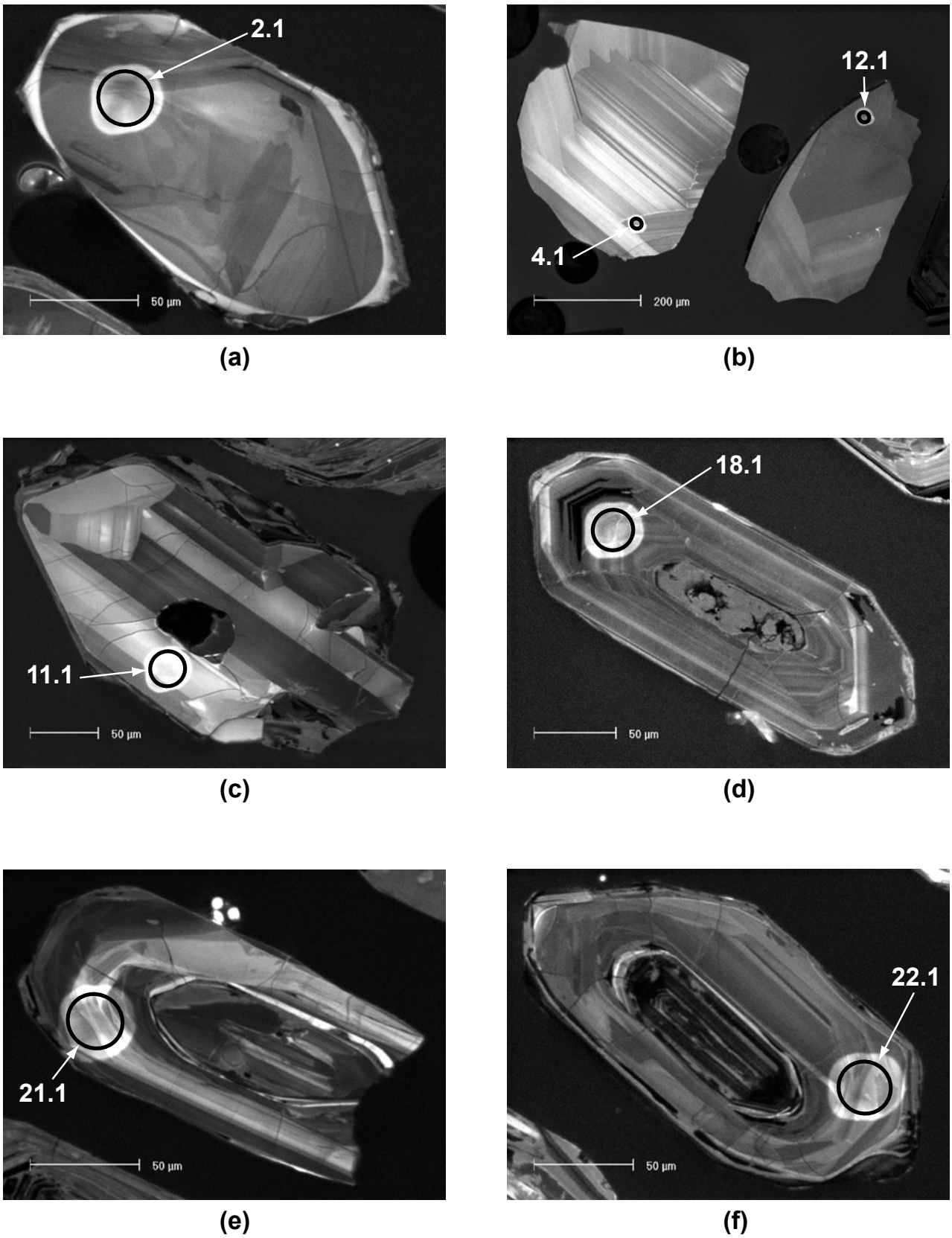

Figure 1. Cathodoluminescence images of representative zircons from sample 178057: biotite monzogranite, Kurrajong Well

Table 1. Ion microprobe analytical results for zircons from sample 178057: biotite monzogranite, Kurrajong Well

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	398	272	286	0.318	0.22653	0.00049	0.18530	0.00074	0.5851	0.0096	18.275	0.308	98	3 028	3
2.1	161	231	111	0.097	0.18038	0.00067	0.39365	0.00144	0.5060	0.0084	12.584	0.219	99	2 656	6
3.1	990	64	99	0.804	0.06806	0.00058	0.02446	0.00123	0.1028	0.0017	0.965	0.019	72	870	18
4.1	43	13	23	0.154	0.18167	0.00153	0.08401	0.00253	0.4930	0.0085	12.349	0.248	97	2 668	14
5.1	89	81	57	0.207	0.18902	0.00103	0.25787	0.00193	0.5127	0.0086	13.361	0.244	98	2 734	9
6.1	32	23	19	0.259	0.18373	0.00182	0.20246	0.00334	0.5070	0.0089	12.844	0.273	98	2 687	16
7.1	127	115	99	0.112	0.23971	0.00083	0.24582	0.00128	0.6042	0.0100	19.968	0.347	98	3 118	6
8.1	58	21	34	0.180	0.18817	0.00126	0.10086	0.00195	0.5244	0.0089	13.606	0.258	100	2 726	11
9.1	156	145	101	0.795	0.18526	0.00088	0.25270	0.00173	0.5096	0.0085	13.018	0.232	98	2 701	8
10.1	366	624	299	0.277	0.20893	0.00048	0.46986	0.00106	0.5602	0.0092	16.139	0.272	99	2 897	4
11.1	62	86	42	0.387	0.17794	0.00130	0.37528	0.00284	0.5046	0.0085	12.380	0.238	100	2 634	12
12.1	76	56	46	0.287	0.18084	0.00110	0.20439	0.00197	0.5044	0.0085	12.578	0.233	99	2 661	10
13.1	105	42	82	0.178	0.26460	0.00093	0.10394	0.00115	0.6572	0.0110	23.977	0.420	99	3 274	6
14.1	142	343	117	0.233	0.18008	0.00082	0.66940	0.00232	0.5134	0.0091	12.747	0.239	101	2 654	8
15.1	108	101	71	0.104	0.18852	0.00078	0.25407	0.00135	0.5240	0.0093	13.621	0.255	100	2 729	7
16.1	121	43	73	0.022	0.19189	0.00076	0.09849	0.00099	0.5410	0.0096	14.313	0.266	101	2 758	7
17.1	315	736	251	0.423	0.17998	0.00053	0.64500	0.00144	0.5025	0.0088	12.471	0.226	99	2 653	5
18.1	344	172	205	1.022	0.17705	0.00063	0.15285	0.00120	0.5040	0.0088	12.303	0.225	100	2 625	6
19.1	192	138	118	0.539	0.17965	0.00069	0.19901	0.00127	0.5076	0.0089	12.573	0.232	100	2 650	6
20.1	246	125	156	1.061	0.18891	0.00072	0.15287	0.00136	0.5300	0.0093	13.806	0.254	100	2 733	6
21.1	90	98	60	0.265	0.18596	0.00099	0.29523	0.00195	0.5226	0.0093	13.399	0.257	100	2 707	9
22.1	228	110	134	0.031	0.17941	0.00051	0.13070	0.00067	0.5194	0.0091	12.848	0.233	102	2 647	5
23.1	141	74	87	0.270	0.18700	0.00072	0.14898	0.00112	0.5330	0.0094	13.743	0.254	101	2 716	6
24.1	455	268	256	0.018	0.17864	0.00036	0.15834	0.00049	0.4892	0.0085	12.049	0.215	97	2 640	3
25.1	446	128	250	0.367	0.18591	0.00044	0.08546	0.00066	0.5060	0.0088	12.971	0.233	98	2 706	4

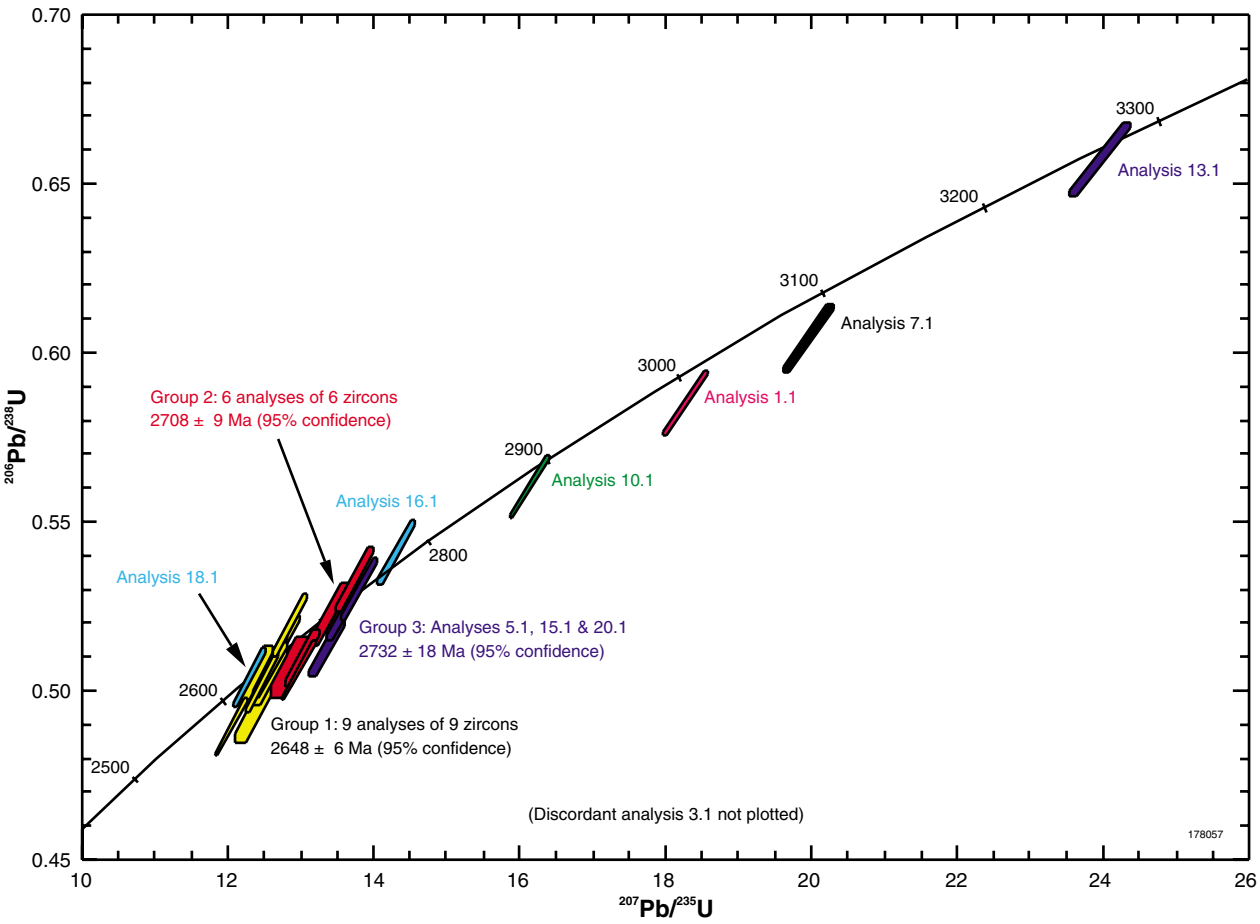

Figure 2. Concordia plot for sample 178057: biotite monzogranite, Kurrajong Well

population and indicating a weighted mean ^{207}Pb/^{206}Pb date of 2732 ± 18 Ma (chi-squared = 0.07). Concordant analysis 18.1 indicates a slightly younger ^{207}Pb/^{206}Pb date, whereas the remaining analyses indicate significantly older ^{207}Pb/^{206}Pb dates than those of Group 1 and cannot be grouped.

The date of 2648 ± 6 Ma indicated by the weighted mean ^{207}Pb/^{206}Pb ratio of nine concordant to slightly discordant analyses of Group 1 is tentatively interpreted as the age of igneous crystallization of the monzogranite. If this interpretation is correct, then discordant analysis 18.1 is interpreted to be of a site that has undergone radiogenic-Pb loss during an ancient disturbance event, whereas the remaining older analyses, including those of Groups 2 and 3, are interpreted to be of xenocryst zircons. An alternative (and less favoured) interpretation is that the analyses of Group 1 were also obtained on xenocryst zircons. If this interpretation is correct, then the ^{207}Pb/^{206}Pb date of 2625 ± 6 Ma (±1σ uncertainty) indicated by concordant analysis 18.1 provides a maximum age for igneous crystallization of the monzogranite.

Recommended reference for this publication:

NELSON, D. R., 2005, 178057: biotite monzogranite, Kurrajong Well; Geochronology dataset 584; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 17/04/2003; Data released: 30/06/2005